Ember Tumuluri

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Education

Northwestern University, Evanston, IL — *August 2026*
Master of Science in Physics

Carnegie Mellon University, Pittsburgh, PA — *May 2024*
Bachelor of Science in Physics/Astrophysics Track

Research Experience

Graduate Research Assistant, Professor Shane Larson — *Evanston, IL*
Northwestern University, CIERA — *September 2024 - August 2026*

- Thesis: Charge Neutralization Timescales of Reissner-Nordstrom Black Holes
- Used the mathematics of general relativity and electromagnetism to create a model to determine a timescale of the neutralization of charged black holes
- Used a combination of algebraic and numerical methods to benchmark the time taken for stellar-mass black holes to lose their electric charge

Research Assistant, Professor Hy Trac — *Pittsburgh, PA*
Carnegie Mellon University, Department of Physics — *August 2023 - May 2024*

- Numerically integrated different rigid-sphere setups to create gravitational force plot to feed into a symbolic regression analysis using Python libraries such as Scipy, PySR, and NumPy
- Created a gravitational profile to more accurately represent gravitational interactions between tracer particles in N-body simulations

Research Assistant, Professor Rupert Croft — *Pittsburgh, PA*
Carnegie Mellon University, Department of Physics — *August 2022 - May 2023*

- Used artificial intelligence to create a density map of the cosmic web within a 3-dimensional region of a cosmological simulation using the Python library TensorFlow
- Created architecture for a convolutional neural network to predict the density of a region of the cosmological simulation from redshift flux from Lyman-α forest

Presentations

36th Midwest Relativity Meeting, Oakland University — *9 October 2026*
Conference Oral Presentation
Title: Timescales of Reissner-Nordstrom Black Hole Charge Neutralization

Physics Undergraduate Research Symposium, Carnegie Mellon University — *20 April 2023*
Poster Presentation
Title: Mapping of the Cosmic Web Using Artificial Intelligence

Projects

The Fate of the Earth, and Other Exoplanets — *Fall 2023*
Astrophysics of Stars and the Galaxy

- Simulated stellar evolution using Modules for Experiments in Stellar Astrophysics (MESA)
- Determined how the habitability zone of a star changes depending on initial conditions such as mass and metallicity
- Created accessible visualization of data using Matplotlib and Numpy to effectively communicate results to fellow students

Finding Dark Matter with Stellar Streams — *Fall 2023*
Astrophysics of Stars and the Galaxy

- Queried data from Gaia's second data release using the Astronomical Data Query Language within the

Astropy package collection

- Identified the stellar stream GD-1 from data, and synthesized sources to argue that the stellar stream's gaps were due to dark matter perturbations

Relevant Work Experience

Graduate Teaching Assistant, Physics I, II, III — *Evanston, IL*
Northwestern University, School of Professional Studies — *September 2025 - June 2026*

- Held office hours once per week to provide guidance to students on homework problems and general concepts within their course
- Graded homework assignments and exams to provide feedback to students on their work done in and out of class
- Proctored exams for students who were given extra exam time to ensure a proper and distraction-free examination space

Undergraduate Teaching Assistant, Physics I — *Pittsburgh, PA*
Carnegie Mellon University, Department of Physics — *August 2021 - December 2021*

- Supervised recitation sessions twice per week to explain concepts to students and assist with practice assignments
- Provided guidance to students with homework problems and clarified concepts to help with studying for exams in a course center once per week

Honors and Awards

Dean's List — *Pittsburgh, PA*
Carnegie Mellon University, Department of Physics — *Fall 2021, Spring 2022, Spring 2023*

Dean's List, High Honors — *Pittsburgh, PA*
Carnegie Mellon University, Department of Physics — *Fall 2023, Spring 2024*

Technologies

Software: Mathematica, LaTeX, Vim, Git, MESA, Numpy, Scipy, COSMIC, Astroquery, Tensorflow

Operating Systems: Mac OS X, Windows

Computer Languages: Python, Fortran, Java, C#, MATLAB, HTML, R, SQL